SHEET 3

1 -

14. What do the following loops print? Work out the answer without using the computer.

A	int s = 1;
	int n;
	for (n = 1; n <= 5; n++)
	{
	s = s + n;
	cout << s;
	}

B	int s = 1;
	int n;
	for (n = 1; n <= 5; cout << s)
	{
	n = n + 2;
	s = s + n;
	}

C	int s = 1;
	int n;
	for (n = 1; n <= 5; n++)
	{
	s = s + n;
	n++;
	}
	cout << s << " " << n;

2 -

2) Write a program that do the following: given a string containing words and numbers, return another string after replacing numbers with words. The given string may contain numbers only from 0 to 9. An example is shown below

Input:

I have 3 Networking books, 0 Database books, and 8 Programming books.

Output:
I have three Networking books, zero Database books, and eight Programming books.

Note: You should use loops, don't use some existing method to replace string.

Hint: when access element in a string make this : Example :

String s = "Hello world";

Char c = s[0]; // access first char in a string and put it in C

Cout << c ; /* at this case C have a value "H" so can use this at previous example to access element and check it */

3) Write a program that returns number of decimal digits in the given input string. E.g.

Input:

2345abc423

Output: //the input contains 7 digits

7

4) Given an integer n, print the following shapes using asterisks and spaces. For example, for n = 5 the following shapes are printed

Input:

5

Output:

```
    *
   **
  ***
 ****
*****
```

5) Given an integer n, print the following shapes using asterisks and spaces. For example, for n = 5 the following shapes are printed

Input:

5

Output:

```
    *
   * *
  * * *
 * * * *
* * * * *
 * * * *
  * * *
   * *
    *
```